

NGUYEN HOANG PHUC

19/10/1996

SUMMARY

About me

Engineering Manager and Solutions Architect with 8+ years - including 3+ years across Prediction, Discriminative AI, and Generative AI - building scalable, distributed systems: microservices, cloud-native infrastructure, IoT and AI-driven platforms for global enterprises. I pair hands-on technical depth with the leadership to grow high-performing teams and turn complex business needs into reliable, production-grade products.

EDUCATION & AWARD

- 2028
|
2026
- **Master's Degree**
Post and Telecommunications Institute of Technology □ Ha Noi, Viet Nam
• Major: Enterprise Information Systems (EIS) - *In Progress*
- 2025
- **MWC Barcelona 2025**
International Exhibition - Mobile World Congress □ Barcelona, Spain
• **Digital Twin** for real-time logistics visibility and predictive analytics - Smart IoT System, Viettel Post
- 2024
- **Viettel Programming Challenge 2024**
Silver medal - VTP Transportation □ Viettel Corporate Campus, Viet Nam
• <https://viettelfamily.com/news/thoi-su/viettel-programming-challenge-tro-lai-voi-giai-thuong-hap-dan>
- **PARCEL+POST EXPO 2024**
International Exhibition - Parcel & Post Industry □ RAI Amsterdam, Netherlands
• **Digital Twin** for automated warehouse operations - robots, forklifts, and RFID-tracked, Viettel Post
- 2018
- **Application of Algorithm Certificate**
Award recipient - international algorithm contest □ SAMSUNG Lab
• Samsung Group
- 2018
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2014
- **Engineer's Degree**
Post and Telecommunications Institute of Technology □ Ha Noi, Viet Nam
• Major: Software Engineering
- 2017
- **SAMSUNG Talent Program**
Selected as 1 of only 13 recipients of the Scholarship from PTIT □ Ha Noi, Viet Nam
• SAMSUNG Vietnam Mobile R&D Center

CONTACT INFO

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Domain:

Artificial Intelligence (AI)
Robotic
IoT, Digital Twins
Cloud Computing
Transportation
Warehousing
Algorithm

System design:

Microservices
Clean Architecture
Cloud Native Application
Edge Computing

Familiar with:

Metaheuristic Algorithm
RAG, Knowledge Graph
AI Guardrails, PII Detection
LiveKit, Whisper, TTS
PyTorch, LightGBM, BentoML,
MLflow
Rule-engine
MongoDB, PostgreSQL, MariaDB
Kafka, Redis
Quarkus, Spring Webflux
Airflow, Time-series DB

Languages:

Java
Python
Dart

English:

Aptis ESOL - B2

EXPERIENCES

Present
|
12/2025

● Vietnam Maritime Commercial Joint Stock Bank (MSB)

Squad Manager (team size: 15)

Project: AI Agentic Platform (AIAP)

Multi-Agent, GenAI, AI Transformation

Stronger come first

AI Agentic Platform:

- My team has built an **AI Guardrail as a Service**: PII detection & redaction, jailbreak filtering, hallucination detection, prompt injection protection
- Long-term **agent persistent memory** via Mem0 - graph + vector hybrid storage (FalkorDB, PostgreSQL pgvector)
- Multi-Agent orchestration with intent-based routing, conversation management, and MCP tool integration
- AI Gateway with semantic routing, semantic caching (vector-similarity dedup), token-aware rate limiting, multi-provider failover, prompt augmentation, and AI observability (cost & latency metrics)
- Context engineering for LLM prompt-cache maximization - stable prefix ordering, conversation **summarization & compaction**, and token-budget optimization to minimize API cost and TTFT
- Graph-based RAG with 6 retrieval modes - local, global, hybrid, mix, naive, and knowledge graph traversal
- Visual Agent Builder Portal - drag-and-drop agent configuration, team workflows, tool & RAG management
- HRIS integration, web search, and extensible MCP-based tool ecosystem
- **VoiceBot** - real-time voice agent using Whisper (speech-to-text) and Text-to-Speech for hands-free conversational banking support
- Stateless refactoring & performance tuning for product-grade adoption

95% jailbreak/PII blocked · 10% semantic-cache hit rate · p95 latency < 2000ms

8% API cost reduction · 80% retrieval recall@k on RAG

11/2025
|
06/2024

● Viettel Post Co., Ltd.

Department Manager (team size: 20)

Project: Transportation Management System (TMS)

Algorithm, Prediction & Discriminative AI, IoT

Stronger come first

Smart Transportation Management System:

- My team has built a Vehicle Routing Problem solver via **Metaheuristic Algorithm** (Tabu Search, Genetic Algorithm) for multi-stop route optimization and load planning
- ETA Prediction - ML regression models (LightGBM, PyTorch LSTM) trained on historical trips, real-time GPS feeds, and traffic patterns to improve on-time delivery accuracy
- Odometer OCR - automated mileage capture using Computer Vision & Deep Learning (YOLOv8, OpenCV, PaddleOCR) to digitize odometer readings for fleet maintenance tracking and trip compliance
- Real-time shipment tracking to streamline logistics and reduce transportation costs

- Data analytics with actionable insights and automated reporting for decision-making
- Customization & deployment for **Laos & Cambodia** branches
11,000+ active users · 8,000+ vehicles tracked · 8K+ trips/day
120 transport partners · 8,000+ hubs/post-offices · 63/34 provinces
24/7 & 99.99% uptime SLA · 200 TPS/instance
ETA MAE < 10 minutes · OCR accuracy 98%

Smart IoT System:

- **Digital Twin** for real-time logistics visibility and predictive analytics - exhibited at **MWC Barcelona 2025**
- Camera AI (DAS) - edge inference with YOLOv8 (detection), MMPose (driver pose/behavior estimation), DeepSORT (multi-object tracking), and TensorRT (inference acceleration) for driver abnormal behavior detection, traffic violation monitoring, fire detection on cargo, real-time safety alerting
- Smart IoT Lock - MQTT protocol, cloud-commanded remote lock/unlock, tamper detection alerts
- GPS Tracking - MQTT protocol, sub-10s refresh rate, geofence triggers & event-driven alerting
- 24/7 Monitoring & Streaming/Playback - WebRTC / HLS / RTSP streams, event-driven recording, live dashboard
5,000+ sensors & other IoT devices · 3,000+ AI-cameras in 1,500 trucks
Real-time monitoring across all logistics hubs
Camera AI detection accuracy 80% · false-alarm rate < 12%

05/2024
|
04/2023

● Viettel Post Co., Ltd.

Tech Leader (team size: 10)

Project: Warehouse Sorting System (Robot, Crossbelt, DWS)

Microservices, Prediction AI, Edge Computing

Stronger come first

Warehouse Sorting System:

- My team has delivered Crossbelt sorters, DWS (dimensioning, weighing, scanning), and **Robotic automation** across all Viettel Post sortation centers
- ETA Prediction - ML regression models (LightGBM, PyTorch LSTM) for sortation dispatch scheduling based on inbound volume and historical throughput
- Edge Computing microservices handling 20,000+ goods per minute
- Dynamic scenario management via Rule-based Engine
- Built core platform libraries covering data access, caching, messaging, security, and observability for all Viettel Post projects
- Customization & deployment for Yody customer via 400 AVG robots
- **Digital Twin** for warehouse ops - 200K+ storage slots, 5 heavy-pallet robots, 5 forklifts, 10 RFID-tracked operators - exhibited at **PARCEL+POST EXPO 2024**, RAI Amsterdam, Netherlands
- Ref: <https://vtv.vn/cong-nghe/ung-dung-cong-nghe-tu-dong-chia-chon-hang-hoa-thong-minh-tai-viet-nam-20240619155209617.htm>
5 Mega SOC · 1M+ parcels/day · 99.5%+ sort accuracy
1,900+ robots AVG · 300+ I/O telescopic lines · 4,000+ crossbelt sorters

24/7 & 99.99% uptime SLA
Dispatch ETA MAE < 5 minutes

03/2023
|
11/2019

● Viettel High Technology Industries Corporation

Backend Development Engineer (fulltime - 3 years)

Project: NFV Management and Orchestration System

Cloud Native, Java, MongoDB

Stronger come first

- Enhanced to the **99.99% uptime** tracking **1M+ data points/second** across 1,000 applications, 200 servers, and 1,000 VMs
- Flexible performance dashboards, automated alerting, and subscription-based reporting
- Clustering across ~50 microservice instances for high availability and distributed consistency
- Compiled all microservices from JIT to AOT (GraalVM), achieving **~50× faster startup** and **~5× lower memory footprint**

11/2021
|
06/2020

● Aspired Inc

Mobile Application Developer (remote - 1 year 6 months)

Project: The Aspired

iOS/Android, Social, E-commerce Application

- Developed cross-platform iOS/Android application on mobile and tablet
- Worked with Jira and Agile/Scrum methodology
- Non-profit project

10/2019
|
09/2018

● SAMSUNG Mobile Vietnam R&D Center

Software Engineer

Project: SAMSUNG MyFiles

Android P/Q/R, File Manager

- Developed SAMSUNG MyFiles Application on Android P/Q/R platform
- Joined via SAMSUNG Talent Program

OTHER CERTIFICATES & AWARDS

05/2026

● Aptis ESOL - English B2

British Council

□ Hanoi, Viet Nam

2024
|
2021

● 4× Advanced Typical Labor

Viettel High Technology Industries Corporation

□ Hanoi, Viet Nam

11/2020

● Mobile Application UI/UX Design
Certificate

□ Capi Creative Agency

02/2018

● TOEIC Certificate

□ IIG Vietnam